Zishuo Zhao

Alibaba Holding

Alibaba Group Holding Limited
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LLM Agents & Reinforcement Learning

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Research Interests

My research focuses on **large language model agents and reinforcement learning**, particularly learning from multi-turn interactions, preference alignment, reasoning, and tool use.

- **Agentic reinforcement learning:** Multi-turn preference learning, turn-level credit assignment, and alignment for LLM agents.
- **Natural language processing:** Source-free domain adaptation for question answering and aspect-based sentiment analysis.
- **Multimodal learning:** Cross-modal retrieval, multimodal graph learning, and vision-language applications.

Education

Sun Yat-sen University *Sep 2021 - Jun 2024*

Master's degree in Intelligent Control

School of Intelligent Systems Engineering, Shenzhen, China

Sun Yat-sen University *Sep 2017 - Jun 2021*

Bachelor's degree in Computational Materials

School of Materials, Guangzhou, China

Professional Experience

Alibaba Holding *Aug 3, 2026 - Present*

Alibaba Group Holding Limited

Alibaba International *Jul 2024 - Jul 3, 2026*

LLM Agents & Reinforcement Learning

- **Customer service agents:** Contributed to AliExpress pre-sales and after-sales agents, including planner-reasoner architectures, tool calling, reward evaluation, and alignment with DPO and GRPO.
- **Multimodal review:** Worked on image-based review with multimodal models, OCR, image classification, and preference alignment.
- **Voice agents:** Worked on Lazada inbound and ICBU outbound voice agents, covering planning, knowledge grounding, interruption handling, and order understanding.

Alibaba International *Dec 2023 - Feb 2024*

Research Intern

Explored RLHF and DPO for customer service language models, reinforcement learning frameworks, and automated model evaluation.

Tencent AI Lab *May 2023 - Oct 2023*

Research Intern

Worked on LLM role-playing, game commentary, and report generation. Participated in research on PPO-based methods for extending model input context.

Publications

Preprints

[1] **Zishuo Zhao**, Kai Chen, Ao Li, Yuan Liu.
Turn-level Multiscale Density Ratio Estimation for LLM Agents.
arXiv:2609.16760, 2026.

[2] Jingyou Xie, Jiayi Kuang, Zhenzhou Lin, Jiarui Ouyang, **Zishuo Zhao**, Ying Shen.
FLEX-CLIP: Feature-Level GEneration Network Enhanced CLIP for X-shot Cross-modal Retrieval.
arXiv:2411.17454, 2024.

Conference Papers

[3] **Zishuo Zhao**, Ziyang Ma, Zhenzhou Lin, Jingyou Xie, Yinghui Li, Ying Shen.
Source-free Domain Adaptation for Aspect-based Sentiment Analysis.
LREC-COLING 2024, pp. 15076-15086.

[4] **Zishuo Zhao**, Yuexiang Xie, Jingyou Xie, Zhenzhou Lin, Yaliang Li, Ying Shen.
Source-Free Unsupervised Domain Adaptation for Question Answering.
IEEE ICASSP 2023, pp. 1-5.

[5] Zhenzhou Lin, **Zishuo Zhao**, Jingyou Xie, Ying Shen.
Hierarchical Type Enhanced Negative Sampling for Knowledge Graph Embedding.
ACM SIGIR 2023, pp. 2047-2051.

[6] Jingyou Xie, **Zishuo Zhao**, Zhenzhou Lin, Ying Shen.
Multimodal Graph Learning for Cross-Modal Retrieval.
SIAM SDM 2023, pp. 145-153.

Technical Skills

Programming and frameworks: Python, verl.
Methods: DPO, GRPO, RLHF, LLM agent development, and multimodal learning.
Languages: English (CET-6, CET-4).